

Care Transition Efficiency & Placement Outcome Analytics

A Process-Based Evaluation of the UAC Program

Research Paper

Submitted to: Unified Mentor (Data Science Internship)

Author: Segabandi Prasanna rani

Year: 2026

ABSTRACT

This study analyzes the efficiency and outcome performance of the Unaccompanied Alien Children (UAC) care pipeline, focusing on the transition from CBP custody to HHS care and eventual sponsor placement. Using a structured dataset, key performance indicators such as transfer efficiency, discharge effectiveness, pipeline throughput, and backlog accumulation were developed. The analysis identifies bottlenecks, temporal trends, and inconsistencies in placement outcomes. A Streamlit-based dashboard was developed to visualize these metrics and support decision-making. The findings highlight areas of inefficiency and provide actionable recommendations to improve care transition speed and reliability.

1. INTRODUCTION

The UAC program operates as a multi-stage pipeline involving:

- CBP custody (entry stage)
- HHS care (processing stage)
- Sponsor placement (exit stage)

Efficiency in this pipeline is critical for:

- reducing delays
- improving child welfare
- optimizing system capacity

However, existing analysis focuses primarily on aggregate counts rather than process efficiency. This study shifts the focus toward flow-based analytics and performance evaluation.

2. PROBLEM STATEMENT

Current monitoring lacks:

- transition efficiency metrics
- discharge effectiveness tracking
- backlog identification
- temporal performance insights

This results in:

- hidden bottlenecks
 - delayed reunification
 - inefficient system operations
-

3. OBJECTIVES

Primary Objectives

- Measure CBP to HHS transfer efficiency
- Evaluate discharge and placement outcomes
- Identify delays and bottlenecks

Secondary Objectives

- Improve reunification timelines
 - Enhance decision-making
 - Support policy-level improvements
-

4. DATASET DESCRIPTION

The dataset consists of daily operational records with the following fields:

- Date
 - Children apprehended and placed in CBP custody
 - Children in CBP custody
 - Children transferred out of CBP custody
 - Children in HHS Care
 - Children discharged from HHS Care
-

5. METHODOLOGY

5.1 Pipeline Modeling

The system is modeled as a process flow:

CBP Custody → HHS Care → Sponsor Placement

5.2 KPI Development

Transfer Efficiency Ratio

Transfers / CBP Custody

Discharge Effectiveness Index

Discharges / HHS Care

Pipeline Throughput Rate

Discharges / Apprehended

Backlog Metrics

CBP Backlog = CBP Custody – Transfers

HHS Backlog = HHS Care – Discharges

Outcome Stability Score

Rolling standard deviation of discharge effectiveness

5.3 Temporal Analysis

- Daily trends
 - Weekday vs weekend analysis
 - Monthly trend analysis
-

5.4 Bottleneck Detection

Threshold-based indicators were used to identify:

- low transfer efficiency
 - low discharge effectiveness
 - high backlog accumulation
-

6. RESULTS AND ANALYSIS

The analysis revealed:

- fluctuations in transfer efficiency over time
 - imbalance between inflow and outflow
 - accumulation of backlog during high intake periods
 - inconsistent discharge performance
-

7. VISUALIZATION

A Streamlit dashboard was developed to visualize:

- pipeline flow trends
 - KPI metrics
 - backlog accumulation
 - efficiency ratios
 - time-based patterns
-

8. KEY INSIGHTS

- sustained high intake without matching discharge increases system pressure
 - backlog growth indicates delays in processing
 - discharge effectiveness varies across time periods
 - bottlenecks occur intermittently rather than continuously
-

9. RECOMMENDATIONS

- improve coordination between CBP and HHS transitions
 - optimize sponsor placement approval process
 - implement real-time monitoring dashboards
 - allocate resources during peak periods
 - introduce automated alerts for inefficiencies
-

10. CONCLUSION

This study demonstrates the importance of evaluating care systems as dynamic pipelines. By introducing efficiency-based metrics and flow analysis, hidden bottlenecks and operational gaps can be identified. The developed dashboard provides a practical tool for improving system performance and decision-making.

11. FUTURE WORK

- integrate machine learning for predictive analysis
 - develop real-time data pipelines
 - include external influencing factors
 - build forecasting models for backlog and outcomes
-